Unified Modeling Language (UML) Reference

RCS 305 ? Software Engineering | Riara University Revision Resource

1. UML Taxonomies

The Unified Modeling Language provides standardized graphical notations across two primary categories: Structural Diagrams (static composition) and Behavioral Diagrams (dynamic execution).

2. Structural Diagrams

? Class Diagrams: Detail classes, attributes, operations, visibility (+public, -private, #protected), and relationships (association, aggregation, composition, generalization/inheritance).
? Component & Deployment Diagrams: Map compiled software packages to physical hardware nodes, network topologies, and container runtimes.

3. Behavioral & Interaction Diagrams

? Use Case Diagrams: Capture functional boundaries, actors, <<include>> dependencies (mandatory sub-flows), and <<extend>> conditions (optional extensions).
? Sequence Diagrams: Model chronologically ordered message passing between object lifelines, asynchronous calls, and activation bars.
? Activity Diagrams: Visualize parallel workflows, decision diamonds, and fork/join concurrency controls.

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